



DESIGN METHODS AND ARCHITECTURAL ISSUES OF INTEGRATED MEDICAL IMAGE DATA BASE SYSTEMS

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Abstract—The past 20 years have seen tremendous changes in medical imaging techniques. New modalities and protocols are expanding the available digital image data at a rapid rate. Yet a framework for gathering, managing, and using multimodal image information in an integrated database environment is missing. The purpose of this paper is to present the experience of implementing an integrated medical image database system at UCSF. We discuss the general system architecture, software design methods, and specific database tools and illustrate them with application examples. Two immediate issues confounding the building of medical image database systems are: lack of supporting infrastructure and inability to index images by content. To circumvent these problems, the evolutionary medical image database system being implemented at UCSF is based on a three-tiered client-server architecture: client medical workstations, database application servers, and a hospital-integrated picture archiving and communication system (HI-PACS). The approach used is to integrate content-based retrieval and knowledge base techniques within the existing HI-PACS to make the whole database system useful in medicine. Copyright © 1996 Elsevier Science Ltd.

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1. INTRODUCTION

Medical images are at the heart of the patient's diagnosis, therapy treatment, surgical planning, and long term follow-up for outcome assessment. Medical imaging is becoming increasingly digital and diverse in scope. In the past two decades, we have witnessed tremendous changes—new techniques include: ultrasound, X-ray computed tomography (CT), magnetic resonance imaging (MRI), digital subtraction angiography (DSA), magnetic source imaging (MSI), positron emission tomography (PET), and single photon emission computed tomography (SPECT), to name just a few.

These digital imaging modalities, currently constitute about 30% of medical imaging examinations of the nation, have revolutionized the means to acquire patient images. They have provided noninvasive means to view anatomical cross-sections and physiological states, and reduced patient radiation dose and examination trauma. The other 70% of imaging examinations on skull, chest, breast, abdomen, and bone are done in conventional X-rays or computed radiography (CR) (1). Different kinds of film digitizers, such as laser scanner, solid-state camera, drum scanner, and video camera, can be used routinely to convert plain X-ray films into digital format for processing.

These advances of imaging technology will certainly continue. However, reorganizations and reengineering of the health care system is shifting the medical imaging focus from the generation and acquisition of images to the post-processing and management of images. The motive is to realize the greatest possible benefit from the data that already exist. The new megachanges of the next 20 years will center around gathering, managing, and using multimedia clinical information.

In the past, medical imaging researchers have been working on individual modalities in isolation while the medical informatics community focused primarily on textual information processing. Thus, in spite of their importance, medical images are poorly incorporated into the overall collection of patient data. Today, tools for archiving, communicating, classifying, organizing, and retrieving medical images to take full advantage of their rich information content are nonexistent (2, 3). This hinders the efficient use of digital modalities in routine medical practice and in improved patient care, such as retrieving multimodal images in real time for use in telemedicine or in virtual reality environments for surgery planning.

Recent technological advances in computers and networks, the press for improved access to health care, the expectation of having health care provided

at lower cost, and the presence of recognized and implemented standards, all bring forth the effort of integrating diverse imaging and data sources into a vertically integrated database system, available for querying and analysis. As the integrated database system expands, it will facilitate the correlation of massive multimedia clinical data to advance the biomedical knowledge. The creation of this framework and the assessment of its performance is considered to be the next important milestone in medical imaging (4). The success of this endeavor will make significant contribution to the national information infrastructure for multimedia health care.

The purpose of this paper is to present the design approach and implementation experience of integrated image database system at the UCSF digital radiology environment. This paper is organized as follows: Section 2 summarizes the new challenging issues of medical image database systems. To address these issues systematically, in Section 3, we describe the design approach and general architecture of the integrated medical image database system at UCSF. Section 4 discusses computing methods and tools that are found to be essential during our development. These application-specific tools are illustrated with implemented examples. Finally, in Section 5, we summarize our presentation and future work.

2. DEFINITIONS AND CHALLENGES

A medical image database system (MIDS) manages a large amount of heterogeneous, changing, pictorial, and symbolic data and provides a means to process, query, and manipulate these medical data in an efficient manner to facilitate optimal decision making in a health care environment. A properly organized imaging database can not only compensate for limitations in human memory and in film recording, but also open up many new vistas to improve patient care, research, teaching, and administration.

Considerable effort has been spent on developing textual medical databases in the past three decades (3). On the other hand, the integration of medical images with textual patient data in database systems has hardly begun. This section summarizes technical issues facing this new form of health care information management. While some of them are common to other image database applications (5–9), the different nature and acquisition of medical images and the legal–medical implications in using medical images for patient care suggest that specific solutions are needed.

2.1. Multimodality

About two decades ago, two-dimensional projectional radiography was the only imaging modality used in clinical examination. Today, a plethora of sectional imaging modalities generate volumetric medical images containing a great variety of previously unattainable biomedical information, e.g. anatomical, geometrical, and spatial, biochemical, physiological, of different body organs. Features and information contained in these images are diverse and interrelated in complex ways that are not always easy to interpret and to represent. Geometric considerations, such as shape and location (9, 10), are as important as organ functionality, as in the heart and liver. Physiological changes, such as the rate of cerebral blood flow, may have subsequent effect on body biochemical activities, such as the concentration of neural metabolites.

2.2. Data heterogeneity

Medical image data are heterogeneous in how they are collected, distributed, and displayed. Images are acquired from scanners of different modalities and in different positions, represented in internal data formats that vary by modality and manufacturer, and displayed in distributed remote workstations with divergent appearance, orientation, and spatial resolution. PET and CT images appear entirely different from one another and are also distinct from other modalities, such as MRI, and ultrasound. Even within the same modality and for the same anatomy, two sets of medical images can vary greatly in slice thickness, dataset orientation, scanning range, and data representation. It is worth noting that diagnostic images are acquired and displayed in gray scale. Thus, color indexing, a popular technique used in other application domains (11), is of little use for biomedical images, except for color microscopy, Doppler ultrasound, and pseudo color enhancement in nuclear medicine modalities.

2.3. Structural and functional contexts

Structural information in a medical image contributes essential knowledge of the disease state and clinical decisions. For example, the location of a tumor and its adjacent anatomical structures have profound implications in treatment planning (spatial context); whereas monitoring of growth or shrinkage of that tumor is an important indicator of the patient's progress in therapy (geometric context). What distinguishes medical images most from other types of digital images, however, is the ability to represent functional information, e.g. biochemistry and physiology, of body parts. As an example,

temporal lobe hypometabolism exhibited in SPECT scans, if not coincident with the tissue atrophy of the suspected region shown in MRI images, may indicate that a different epileptic focus as otherwise would be concluded.

2.4. Imprecision

Images contain information that is often imprecise and can only be expressed informally. This phenomenon applies to entire data sets as well as features within individual images. The boundary of a tumor, for example, may be fuzzy, depending on the resolution of the imaging modality that is used. Yet, the accurate delineation of a tumor is an essential first step for curative therapy. Worst, the imprecision of image resolution is often compounded by the vague description of features extracted from the images or ambiguous coding of disease classification in diagnostic reports. For instance, the diagnosis "acute myocardial infarction, anterior wall" imparts a sense of certainty and specificity, and can be linked to ICD-9 (International Classification of Diseases, ninth revision) code 410.10. On the other hand, for the diagnosis "left ventricular aneurysm", an agreeable definition has yet to reach within the medical community. New, more expressive medical data dictionary and representation models that can cope with such imprecision are needed.

2.5. Temporal

Staging of the disease state and the monitoring of patient progress over time are fundamental to diagnostic and therapeutic decisions and to outcome assessment in long term follow up. The ability to define and track temporal relation in the medical image sets of a patient taken at different periods, together with the medical history of that patient, is an essential component of medical image databases. As the database expands to incorporate a large mass of similar patient data, tools developed for the intra-subject temporal monitoring can also be extended to inter-subject temporal comparison to improve prognostic and diagnostic outcomes.

2.6. Large datasets

Like other imaging applications, medical images are distinct from textual records in their large data sizes. Images acquired in one examination can range from 1 or 2 megabytes in nuclear medicine to 32 megabytes or more in mammograms and digital radiographs. Digital medical images accumulated in a 1000 bed hospital are estimated on the order of a terabyte a year (12). These imaging modalities generate such a large volume of image data that

traditional methods for information management are no longer adequate. More advanced tools and schemes to manage multimodal images and their complementary textual information for database processing are needed.

2.7. Infrastructure support

The accessibility to heterogeneous images, patient data, and medical reference materials from diverse sources in the clinical decision making process presupposes the existence of a communication infrastructure. The reality, however, is that most databases and information systems in hospitals today are disjoint, under the control of different medical sections and departments, and often do not communicate with external reference database networks well. The isolation creates many technological and administrative barriers that make the gathering of information cohesively from different medical images extremely difficult (2, 13). This lack of supporting infrastructure causes the concept of multimedia data integration far from practical deployment.

2.8. Security

Along with the multimedia data integration and communication comes a new issue: How do we establish integrity and privacy in medical records that exist only in the easily altered digital storage of computers. This issue is especially prominent when an image database framework is connecting to national and international health care networks. An open yet insecure system is unlikely to be acceptable by the clinicians due to legal-medical complications. The existing means of security measures, such as checking for personal ID or authorization password, enforcing administrative policy, and prioritizing user privileges, all have known problems in the ubiquitous networking environment (14, 15).

3. UCSF IMAGE DATABASE SYSTEM ARCHITECTURE

Among the new issues confounding the building of medical image database systems, two fundamental ones stand out: lack of supporting infrastructure and inability to index information by image features. These two issues form the core of our research of integrated medical image database system (MIDS). A MIDS is a repository of integrated information, i.e. various kinds of medical images and clinical text, available for querying and analysis. As relevant information becomes available or is updated, the information is extracted from its source, translated into a common data model, and integrated with

existing data at the MIDS. Within this integrated medical system, queries can be answered and data analysis can be performed quickly and efficiently since the information is directly available, with model and semantic differences already resolved. Furthermore, MIDS data can be accessed without slowing down the operations of original information sources, such as a radiology information system or an MR imaging system, and is available even when these information sources are inaccessible.

The concept of integrated MIDS is similar to the notion of the data warehouse discussed in general database literature (16, 17). The advantage of radiologic imaging is that the raw image and patient reports, once generated, are rarely modified. The update is limited to a subset of processed and extracted data for decision support queries. Hence, MIDS avoids much of the difficulty in providing a consistent database source compared to business or engineering applications. In radiology, this information framework is a logical progression from the primary digital modalities and the stand-alone medical information systems towards the networked multimedia database systems. Building a comprehensive, enterprise framework that encompasses generation, acquisition, processing, management, and communication of radiologic images is an evolutionary process. System design must provide the capability to phase out old components and incorporate new devices and technology as the system evolves.

3.1. Hospital integrated picture archiving and communication system

The MIDS being developed at UCSF is based on a three-tiered client-server architecture. This architecture divides applications into parts that run on different types of computers: a picture archiving and communication system (PACS), an image database server (IDBS), and client workstations for specific medical applications. A PACS is a system integration of many components, including image acquisition devices, computers, networks, display stations, medical databases, and legacy information systems. Figure 1 shows the hospital-integrated picture archiving and communication system (HI-PACS) implemented at UCSF. The system contains a centralized, dynamic, and massive storage of multimedia clinical data acquired from various imaging scanners and databases across the UC hospitals.

The implementation of this HI-PACS emphasizes recognized and implemented standards, open systems connectivity, hierarchy memory management, database integration, and security. The UCSF PACS connects not only various imaging

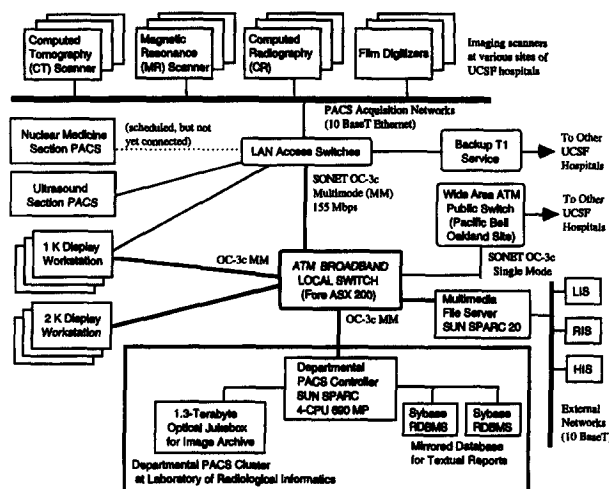


Fig. 1. Schematic diagram of the hospital integrated PACS at UCSF.

scanners directly, but also smaller scale PACS of subspecialty sections and textual information sources, such as hospital information system (HIS), radiology information system (RIS), and library information system (LIS). The PACS fiber-optic backbone has been installed interconnecting four major facilities in UCSF, i.e. Moffitt and Long hospitals, ambulatory care clinic, and campus library, to provide a data warehouse integrating information from multiple imaging and data sources.

Wide area connections between UCSF and affiliated hospitals in the San Francisco Bay area using asynchronous transmission mode (ATM—OC-3c, 155 Megabits per second (Mbps) for real-time telemedicine) are being implemented. Image communications between UCSF hospitals using frame relay technology (T1 link—1.54 Mbps) for non real time telemanagement have been in operation. The data warehouse at the departmental PACS cluster can store up to 1.3 terabytes of image and text data. These PACS data can be retrieved and transmitted over the networks to high resolution, multi-monitor display stations located in intensive care units and radiology reading rooms for clinical review.

The intent of picture archiving and communication systems is to provide management and fast review of vast volumes of image and text files in a digital radiology department. The PACS file management system can only handle a query by artificial keys, such as patient name or a hospital ID. It lacks the means to classifying and indexing these data files by their information contents. Thus much of the rich, useful patient information stored in PACS has been

not tapped for broader medical practice, research, and teaching.

To provide genuine database management capabilities, we develop an image database server that can extract relevant features from PACS images and textual reports and organize the extracted data, together with annotated image slices, into a common data model. The image database server can also process various computer-aided medical applications, such as fast indexing of medical records, interactive visualization of images, and activate rules for decision making. It consists of a set of sharable, multitasking components that interact with client workstations and PACS. Section 3.2 discusses this service tier in more detail.

The client workstation provides a graphical user interface (GUI) to display multimedia information and communicate with the image database server for specific biomedical application processing, such as computer-aided diagnosis and surgical planning (18, 19). Screens and GUI routines are managed in a repository as objects to permit a modular approach to development and foster reuse of objects. In addition, tools that are not necessary for traditional textual databases but are perceived as essential will then be developed and integrated into the overall framework. This three-tiered system development is a bottom up endeavor, progressing from building the rudimentary level of image acquisition, archiving, and communication to providing sophisticated image database management. It lets the developers and system administrators to separate the specific application logic from general processing logic. This modularity allows new services and clinical changes to be more rapidly incorporated into client applications.

3.2. Image database server

The image database server (IDBS) is a centrally located core of software objects or application programs for accessing, processing, managing, and manipulating data from distributed information sources in the HI-PACS environment as if they were from a single database. This middleware database server, resided in between HI-PACS and client medical workstations, consists of three cooperating processes or engines, running in parallel. It also serves to filter all client requests through various, centrally controlled levels of security, such as IP (Internet Protocol) firewalls, access control, and image authentication. Figure 2 shows the key components in our medical image database framework.

1. **Database Engine** that contains a collection of objects and a set of operations that they can perform. Data models are devised to classify, organize, and represent images and text to support information retrieval by image content for various biomedical applications. The database engine treats system components and multimedia data of the entire database framework as a logical conglomeration of distributed, interacting objects of various levels of granularity. For rapid prototyping purpose, we decided to use a commercially available relational database management system and then to develop customized object software modules on top of the commercial system for special applications. The current database engine used is based on the object relational technology—Sybase database management system (Sybase Inc., Emeryville, CA) for the underlying persistent data storage.
2. **Image Processing Engine** that provides image processing services for various medical applications. These services include computationally intensive operations, such as image registration and feature extraction. The decision to locate heavy duty imaging application routines in the powerful image database server is to reduce the hardware and software requirements on the client workstations. The medical image processing and visualization software packages used include VIDA (University of Iowa College of Medicine, Iowa City, IA), 3DVIEWNIX (Hospital of the University of Pennsylvania, Philadelphia, PA), and several in-house developed programs (18–22).
3. **Knowledge Base Engine** that consists of three kinds of knowledge base modules. Rule-based

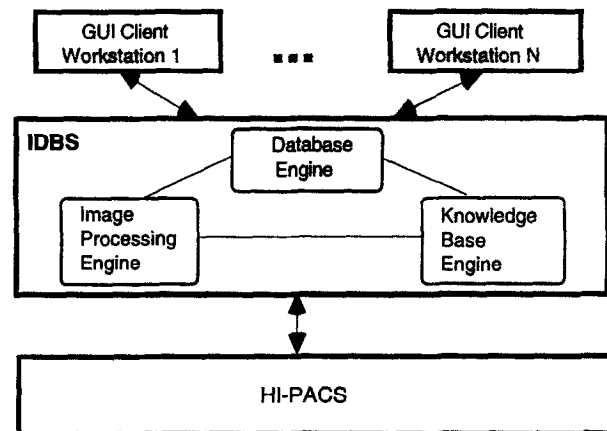


Fig. 2. The medical image base system (MIDS) architecture with IDBS over HI-PACS.

3. **Knowledge Base Engine** that consists of three kinds of knowledge base modules. Rule-based modules that encode general and specific medical rules for decision making, case-based modules that aid the navigation of database through similar or reference imaging cases, and model-based modules that provide interactive work-up models of different disease categories closely involving physicians making joint decisions. Our current knowledge base engine includes a mix of C routines, embedded SQL (Structured Query Language) statements, and GEL (Gain Extension Language) scripts of Gain Momentum.
4. **GUI Client Workstations** each provides a graphical user interface (GUI) for the local user to request applications of the image database server to perform information retrieval, manipulation, analysis, and update. Each client workstation also contains simple image processing capabilities, such as thresholding and filtering, but accesses the IDBS for more sophisticated database services. Our current database front-end development tools include Gain Momentum and several Web browsers, such as Mosaic and Netscape. Gain Momentum is a Motif (UNIX) application development environment with an underlying object-oriented database (Gain Technology, Sybase Inc., Palo Alto, CA). The underlying object-oriented database contains the objects that allow the client applications to access, manipulate, or present the underlying multimedia data in the server. Web browsers are efficient hypermedia navigation tools for "public domain" file documents. To allow complex, hierarchical representation of database information, however, we have to incorporate new software tools with extensive multimedia querying capabilities into the web browsers. Due to the complex programming effort involved, our current web-based database application is limited to simple teaching file queries (23).
5. **HI-PACS** consists of the PACS multimedia file storehouse and a federation of distributed, heterogeneous medical databases, imaging acquisition devices, and hospital information systems, providing the environment in which the above system components run.

The IDBS prototype is being developed on a 4-processor SUN 690 MP SPARC computer. Security mechanisms are set up to allow the database server to access the multimedia information of HI-PACS with fine-grained control over individual transactions and

data items, yet without running the risk of contaminating daily clinical services.

Figure 2 provides a generic architecture of integrated medical image database system. Since medical technology research is driven by well-defined hypotheses, our development focuses on particular clinical applications that have specific kinds of representation and queries applied to medical images. The applications currently under development are: surgical planning of epilepsy (19), neurological diagnosis of dysmyelination and multiple sclerosis, and the age assessment of hand bone (21). By restricting the scope of applications and the types of images, we will provide some useful though initially limited functionality for content-based retrieval of medical images. The formation of these medical image databases follows knowledge and feature-based indexing methods (11, 24), but adds sophisticated services in automatic image registration, image feature extraction, multimodal image presentation, and multimedia client server application development in HI-PACS environment.

4. MEDICAL IMAGING DATABASE TOOLS

Section 3 describes the general architecture and design of our integrated image information system. This section discusses specific tools being developed within this framework for various medical applications. The key idea behind the MIDS approach is to extract, filter, and integrate relevant information in advance of queries. Extraction of keywords and phrases from textual reports into medical databases can be done automatically via string pattern recognition. On the other hand, current medical image databases allow queries to be formed based on the qualitative features of images that have been manually annotated.

For content-based information query, the next generation of medical image databases must have capabilities to recognize and quantitate image content and merge the quantitated image data with textual patient data into a common data model. The content of a medical image includes three levels of abstraction: (i) features or attributes, such as the texture pattern of an organ, the shape of a breast tumor, and the signal strength of brain metabolites; (ii) structural objects, such as image segments corresponding to anatomical regions; and (iii) semantic information, such as the types of relationships between two objects.

Figure 3 illustrates the operational steps that the MIDS taken to extract image and textual features from the on line PACS data. The feature extraction is

based on the a priori approach, rather than the dynamic and automatic feature extraction during user query as proposed in many non-medical image database applications (6, 7).

The extraction and composition of textual data from the diagnostic reports and the DICOM (Digital Imaging and Communications in Medicine) header fields can be automatic, whereas the segmentation and extraction of medical images often are done interactively. Further, we divided the image features into primitive and logical. Primitive features are directly obtained from the medical images, such as volume, shape, and texture of certain organs in CT images or metabolic activities of brain tissue in PET scans. Logical features are abstract representations of images at various levels of detail and deeper domain semantics. For example, whether the volume of an anatomic structure is normal or whether certain brain tissue is hypometabolic in reference to certain established data. These logical features are synthesized from primitive ones and additional domain knowledge. All features extracted are entered into appropriate attributes of the objects defined in a data model to facilitate subsequent information query by content.

In what follows, we describe the application-specific tools that are found to be essential in the MIDS development. Our emphasis at this stage will be on those which have direct impact on the building of a working prototype, rather than on query

performance and system administration. Few of these tools are currently supported in commercial systems, and many of them are still awaiting efficient solutions from the database community. In addition, examples are taken from our applications to better illustrate the methods we used, as well as to highlight the unresolved problem areas.

4.1. Image data modeling

An image database system must be able to transform pixel based representation of medical images and their contents into object based representation for symbolic manipulation. This transformation is a crucial step for building any content-based pictorial query system.

Once medical images can be efficiently represented in a content-based manner, by objects rather than pixels, then a wealth of established database theory, methods, and tools can be readily applied for content-based access, creation, and manipulation. In addition, clinical decisions are based on not only information obtained from medical images, but also from textual and graphical data of the patients, including medical history, physical examination, laboratory tests, diagnostic workups, and pathological results. These images and data from diverse sources must be integrated with the physicians' medical knowledge and available reference materials to deal with therapeutic or diagnostic decision at hand. More expressive data models than those used in

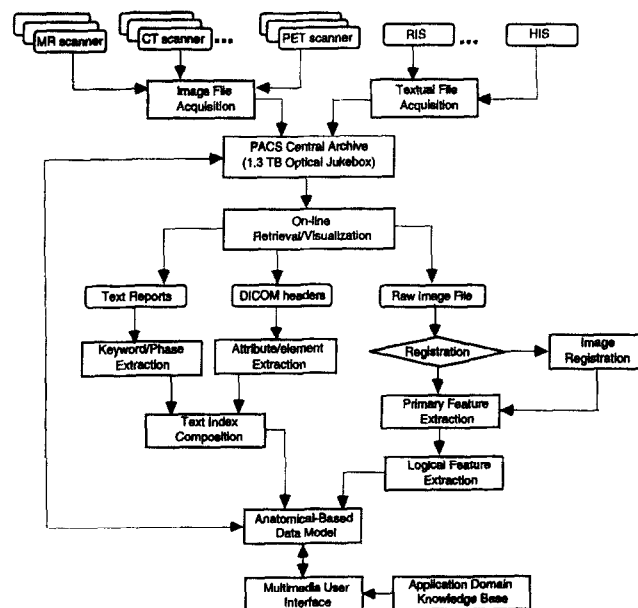


Fig. 3. The operational flow of extracting image and text features into an anatomical based data model for subsequent content-based image indexing in MIDS. Specific knowledge or heuristics is triggered to aid the query and navigation of the medical image database.

textual databases are needed in modeling and indexing such multimedia medical data. Relational approach is perceived to be ineffective in modeling multimedia medical data. We have taken the object oriented approach to represent anatomical structures.

Figure 4 shows an object data model for hand bone structures (21, 24) and a computed radiography (CR) image instance of left hand in posterior-anterior (PA) view. Often, textual data indices are extracted automatically from associated patient reports to complement the descriptors of image object features.

Object data models also provide flexible means of indexing and linking both images and text and can incorporate biomedical production rules for decision aids. Let us consider the following query for determining bone normality of hand computed radiographs.

(Q1) Assess the skeletal age of a pediatric patient based on the given CR left hand image and decide whether the skeletal maturity is normal.

The assessment of skeletal maturity is an important subject in pediatric radiology. Database solution to such a problem is thus of significant clinical value. This query, however, is vague; only the information about the patient's CR hand images is given and 'normal' is an ambiguous term in medicine. A computerized method has been developed in our research (21) to assess bone age, based on measurements from different structure of the hand bone. This method consists of the following steps: (1) separate the third finger image from the given hand image pattern; (2) measure the lengths of the distal, middle, and proximal phalanx; (3) convert the measurement into age using the standard phalangeal length table in medicine; and (4) compare the estimated age with the patient's age. The skeletal maturity is judged to be normal if the patient's age is within the limit of standard deviation.

These computational steps can be modeled as production rules in third finger object class of the skeletal model as shown in Fig. 4. The image database system thus directs an ambiguous query of the type similar to query (Q1) to the third finger object. If there is no pre-calculated value in the patient finger object's attributes, that object instance would invoke the encapsulated production rules to extract the features and analyze the results.

The hand bone application is an example of content-based medical image database using a single imaging modality. The tools and data models developed, though optimized for this specific research, are restricted because they are derived

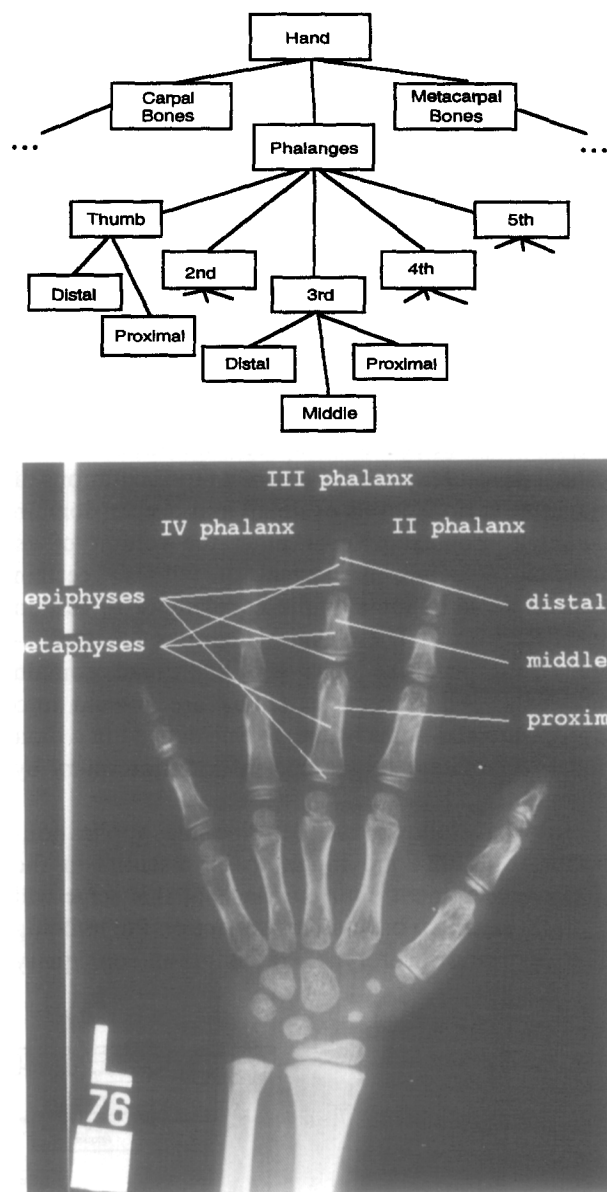


Fig. 4. (a) The object-oriented, anatomic data model of hand bone structure. (b) Digital representation of hand bone structure in a computed radiograph (courtesy of Dr Ewa Pietka).

from digital radiography only. Daily medical decisions often desire the gathering and correlation of information from several types of patient images as well as from various data sources. In addition, the correlation of multimodal images of the same body organ may be more effectively modeled by biomedical relationships between objects of different modalities. Object-relational approach, such as those advocated in UniSQLTM (UniSQL Inc., Austin, Texas) and IllustraTM (Illustra Technology, Oakland, CA), may offer alternate solutions in

multimodal image modeling than pure object oriented approach.

4.2. Image fusion

If the pixel-object transformation is the crucial step for realizing content-based representation, the fusion of volumetric image pairs is the key to enabling multimodal image databases.

The fusion of images consists of two major steps: (i) the accurate registration of images from different kinds of imaging modalities, different periods of the same patient, or different patients; and (ii) the correlation and extraction of biomedical information represented in registered images to derive new, previous unavailable medical knowledge.

4.2.1. *Medical image registration.* Figure 5 provides a case of the use of image registration to align two sets of MRI images taken from a 30 yr old patient suffering multiple sclerosis; a disease condition marked by patches of hardened tissue in the brain or the spinal cord and associated especially with partial or complete paralysis and jerking muscle tremor. The top row shows the patient's brain scan taken a year ago while the middle row is the recent one. The objective is to look for differences between the two scans to evaluate changes related to disease progression (marked by the sizes of high intensity dark regions around the ventricular regions of the brain). Using conventional films, the radiologist has to rely on imagination to correct for discrepancies in head position that result from different positioning in

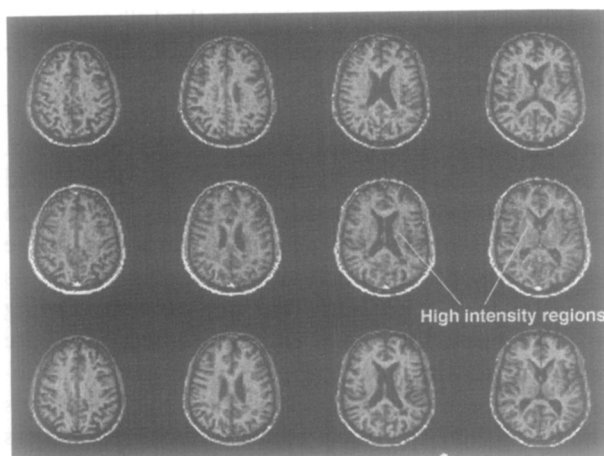


Fig. 5. Automated Image registration of two MRI image datasets of a multiple sclerosis patient. Dark X-shaped regions in the center of the brain are left and right ventricles. Top: MR scans taken 1 yr ago; middle: recent MR scans of different orientation and position; bottom: registered image of the top row with respect to the second row.

the two scans. Even so, he or she cannot compare changes of lesions in top scan to middle scan.

The IDBS provides automatic image retrieval and registration services for neuro images. A client neurological workstation requests the database server to retrieve the specified image pair stored in PACS in real time and automatically align the first image based on its 3D dataset with the recent one. The result is displayed in the bottom row, which indicates that the disease condition of the patient has not worsened, as the size of high intensity regions, i.e. lesions, around the ventricular areas remain the same. This example also illustrates the use of the temporal relation of medical images in monitoring the patient's progress or response to treatment.

We adopted the clinically validated Woods' algorithms for automatic image registration (25) and since then have extended the registration routines in the image processing engine to cover the inter- and intra-registration of four different kinds of brain images, i.e., MRI, PET, magnetic resonance spectroscopy (MRS), and magnetoencephalography (MEG) (19).

Automated image registration with on-line access to a vast mass of image data makes possible many value-added functions that cannot be done in the conventional medical record, such as looking for subtle changes in signal across time, combining different types of functional (such as PET and SPECT images) and structural information (such as CT and MRI images), and comparing different subjects or patients to one another for better clinical decision making. What is more important, it prepares the stage for the second step of image fusion, i.e. providing a means of correlating and quantitating content information of different images.

4.2.2. Image correlation and features extraction.

The correlated image datasets provide diagnostic insights not readily apparent when examining the imaging modalities independently. They are encoded into the targeted data model to serve as definitive indexing in image query. For instance, registration of functional images obtained using nuclear medicine modalities, such as PET or SPECT, with anatomic MRI images is of particular importance in neurological diagnosis, since it allows the intrinsically better spatial resolution of the MRI image to be used in interpreting the functional activities of the brain.

Figure 6 illustrates such a case of MRI defined regions of interest (ROI) registered with PET. In this image the left and right amygdala are delineated. The clear definition of anatomical volumes for MRI allows quantitative analysis of functional informa-

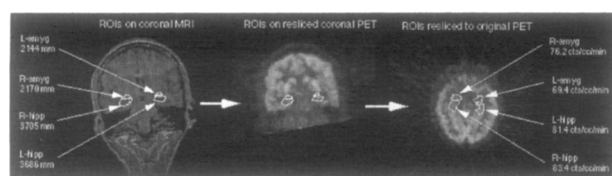


Fig. 6. Mapping regions of interest from MRI to PET. For simplicity, we only show a two-dimensional case here. The middle one shows the aligned position of PET image with that of MRI image while the right one is the orientation of original acquired PET image. The outlined areas, right and left amygdala and hippocampus, are the regions of interest of all three images in the common space.

tion (metabolic count of glucose consumption) captured in PET scans. Various methods for specifying and manipulating three-dimensional regions for quantitative analysis of functional brain images, such as PET, MRS, and MEG, based on the structural MRI images have been implemented in our MIDS testbed (16, 18).

The extracted quantitative data will be important to understand the clinical implementation of a particular therapeutic or diagnostic decision while providing exact numerical indices for ad hoc queries by image features, e.g. MRI anatomical volume or PET glucose uptake count. Similar structural mapping with other functional images, such as the concentration of energy metabolites measured in magnetic resonance spectroscopy, will reveal a different aspect of biomedical information and will serve yet another kind of indexing for image queries.

Figure 7 shows a multimedia user interface developed in our client neurological workstation. This user interface enables real time retrieval of PACS images for comparison and reference during diagnostic workup of epilepsy surgery candidates. Query of the image database can be by image content, e.g. MRI hippocampal volume or PET glucose concentration, at the lower left window, by patient attributes, e.g., name, age, and sex, at the upper left window, or a combination of these features. The sliders in the "Query by Image Attributes" window can be used to specify the range of the image attributes for data retrieval. The neuroimaging database returns with a list of seven patients satisfying the combined image and patient attribute constraints. The user can select any patient name to retrieve and display MRI and PET images (in this case, the third patient). By default, only two pairs of representative, co-registered image slices are displayed. Once the images of interest are retrieved, the user can invoke the 3D image visualization routines of the image database server for more sophisticated image rendering and manipulation.

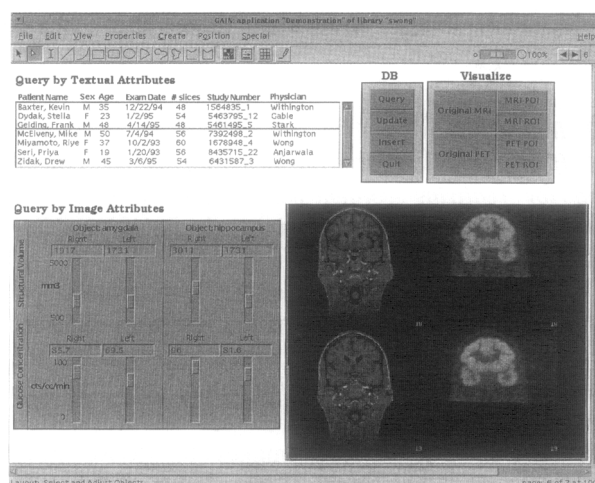
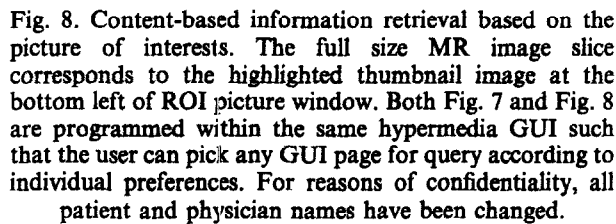


Fig. 7. A content-based GUI developed for retrieving patient PET and MRI images for reference and comparison during surgical planning and diagnostic workup of a new epilepsy case. For simplicity, MRS and MEG image features of anatomical objects are not shown. The images shown in the right are the third patient in the list selected by the clinician. For reasons of confidentiality all patient and physician names have been changed.

Figure 8 shows yet another kind of content-based information query. The user first specifies the structural, functional, and textual attributes of the MRI studies of interest. The image database server returns with a list of patients satisfying the query constraints, and, in addition, a set of representative picture of interests of these patient cases in thumbnail form. The user then clicks on the particular thumbnail image to zoom it up to the full size or to directly retrieve the complete 3D MR dataset for further study.

Image fusion being an integrated part of medical image databases is just in the beginning. Tools for rapid implementation of automated image registration procedures are still lacking. In particular, tools for the accurate, automatic image registration of non-rigid body organs, such as prostate gland and lung, are non-existent. Owing to this limitation, our current applications are restricted to rigid body organs or organ encapsulated by non-moving solid boundary, such as hands and brain. On the other hand, medical researchers are still trying to understand many biomedical implications and opportunities of the new, powerful information derived in the image fusion process. The establishment of medical image databases for large scale statistical analysis or 'data mining' will in turn contribute to the advances of biomedical knowledge and clinical practice.



Pixel-object transformation and multimodal image fusion form the foundation of content-based image retrieval, but they must be integrated with case-based reasoning techniques for medical decision making to make the entire image database systems useful in medicine.

On the other hand, the user can develop possible hypotheses to explore and navigate through the image database. Thus, general tools that allow for "show me one like this shape but smaller" or "find me something of size in between the two" will provide powerful means of synthesizing new concepts and knowledge. This "changing the field of view"

(Q2) Find all MR head examinations of patients with multiple sclerosis which show white spots in the MR scans, categorize all the white spots found by size and locations.

MR images of T2-weighted sequences are normally used to identify white spots or fiber deposits in the brains of multiple sclerosis patients. The white spots are usually found within white matter, particularly the periventricular area around the lateral ventricles, the optic nerves, brain stem, and spinal cord. This application requires the extraction and quantitation of white spot volumes in these brain regions.

4.4. Customized schema and languages

In DBMS the schema represents the information model of the application. In object oriented databases the schema represents the various classes, the

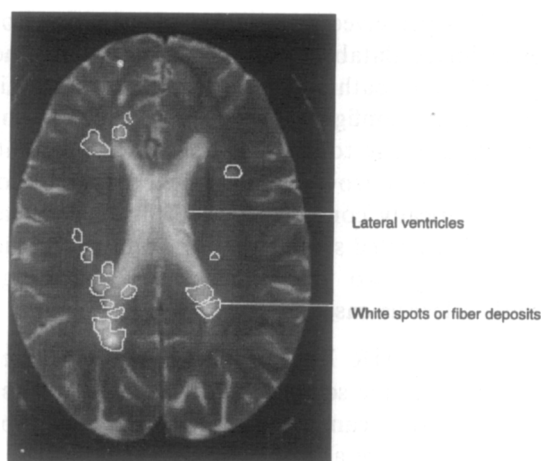


Fig. 9. A cross-sectional slice of a T2 weighted brain MR images of a 40 yr old multiple sclerosis patient. The X-shaped bright regions in the center brain are the left and right lateral ventricles. The objects of interest, i.e. white spots around the periventricular area can be segmented out interactively and their sizes can be quantified by using the image processing engine of IDBS.

attributes (fields, instance variables), and the relationships between the classes. In relational databases the schema defines the tables, both base and derived; various integrity constraints on the tables; and other meta-information of the database extensions. The schema therefore represents the structure of the information such that a data manipulation language can be used to query the information stored.

Database languages for multimedia applications has been discussed widely in the literature (6–9). Here we highlight some of the schema issues in medical imaging domain. Clinicians view and use the same medical image database in a vastly different manner. For example, a radiation oncologist may ask the database system about the physical shape and spatial position of tumor regions targeted for therapy, whereas the diagnostician may query the functional and morphological features of the tumors for staging of the tumor. In general, image queries in medicine can be categorized as follows:

- *Clinical practice* concerns the daily delivery of health care such as diagnosis and therapy. Some typical image queries include: (1) retrieve the current imaging studies and/or previous imaging studies (may be of different modalities) of a patient; (2) retrieve all imaging studies which show common (or different) image features of the current case; and (3) inquire about biomedical attributes of an object or a group of objects of interests, such as size of a lung nodule and glucose concentration in a brain region.

- *Education* concerns the teaching of graduate and medical students, residents, and fellows. Some desired query capabilities include: (1) retrieve pathologically proven medical images according to a particular type of disease; (2) step through a sequence of examination questions and medical images dynamically based on the user response to the questions; and (3) ask the image database to provide another set of example images by modifying certain descriptors, such as severity of disease and age of patient population, of a teaching case.
- *Research* uses databases to support large scale, systematic study of diseases and medical procedures. Typical queries involve mass screening a large number of clinical imaging studies from various patient groups, i.e. sex, age, race, and medical history. Given the size and region of a brain tumor, for example, queries may be issued to provide statistics on the improvement of patients operated under various treatment methods. On the other hand, a hypothesis (e.g. a certain drug is effective in controlling multiple sclerosis) can be composed and then be validated by a series of queries sent to the database system (e.g. measure the variation in the sizes and numbers of white spots shown in MR head images of all multiple sclerosis patients). Such queries lead to new biomedical knowledge.

Thus, the schema of an integrated database system should be able to evolve over time to serve the expanding scope of clinical applications. Management tools should be provided to cope with such changes. The description of a query language will be dependent on the underlying data model used. Standard query languages are encouraged to ensure the interoperability of medical image databases with medical information systems using similar data models, e.g. SQL (structured query language) for relational DBMS and OQL (object query language) for object oriented DBMS. Iconic or pictorial queries built on top of these standard languages are essential to reap the full benefit of medical image databases, and this is the direction of our GUI development effort. The use of object icons and graphical relations provides an efficient means to develop customized queries and better conceptualization for the problem at hand.

4.5. Image manipulation and communication

A medical image database system must incorporate image processing functions into its symbolic processing capabilities so that, ultimately, image

database engines manage the data or information stored and manipulated by these application tools.

Unless image database systems can acquire and manipulate image pixels as easy as they have done for alphanumeric symbols, the systems developed will not be of any practical use. These new functions include: (1) manipulate multimodal images, such as zoom, pan, rotation, contrast enhancement, region of interest contours; (2) recognize and segment images, such as edge detection, morphologically filtering, and feature extraction; (3) visualize images, such as surface and volume rendering, texture discrimination, mosaic, cine, and multiple and multimodal image display; and (4) communicate images, such as compression for speedy transmission and real time protocols.

Simple image processing tools, such as edge enhancement and filtering, used in our client medical workstations are inherited from the existing PACS display stations. More computational intensive tasks, such as visualization and volume rendering, are delegated to the image processing engine of the IDB server. However, existing image segmentation techniques become useless when the boundary between the object of interest and the background is fuzzy. The segmentation of the brain objects in our applications is being done interactively. This constitutes the most time-consuming operation in our development. Efficient tools for automatic feature recognition of fuzzy images, image registration of non rigid objects, and real time transmission of volumetric image datasets over broadband networks are particularly lacking.

4.6. Data integrity and privacy

Integrity and privacy of electronic medical records in the integrated image information environment can be divided into the database server level and file transfer level.

PACS file servers can provide data security, but at the file and directory level. This means various groups and users can be given or denied access to certain patient files or directories. Some directories might not be "visible" to certain users. Image database servers, on the other hand, can provide security at the object level. Users can be granted various privileges on logical data objects and collections of objects. The granted privilege is also more diverse in scope. It might be for read or update operations, create operations, or execution of methods associated with persistent objects. With a file-server architecture such as PACS, the PACS controller software checks and imposes the file- and directory-level security that executes on the PACS

controller node. With the MIDS database-server architecture, the software that stores and executes database-server security resides and is executed on the middleware image database server node. This provides more flexibility and fine grain control on the different levels of user-access privilege.

Another level of security concerns the transmission of queried image data to the requested client workstation via the local or wide area networks. Cryptography is considered as one of the strongest methods to ensure data security and authenticity. MIDS can provide software encryption using the public-key algorithms and a combination of public- and private-key algorithms for data privacy. The key length used ranged from 256 to 1024 bits. Software encryption, however, adds considerable computation time for 3D image datasets or computed radiographs (22). It thus does not support real time delivery of medical images. For example, encrypting a volumetric brain MR image of 8 Mbytes takes about 2 min in a Sun SPARC LX workstation (transmission time is not a factor with our 155 Mbps ATM links). Nevertheless, encrypting a few selected MR image slices ($256 \times 256 \times 2$ bytes/slice) for the referring physician takes over 1 or 2 s. The latter thus is amenable for real time clinical practice.

In certain clinical scenarios, such as in emergency room or during surgery operation, the integrated information framework should provide real time transmission of full volume datasets as well as the checking of data integrity or authenticity (22). One approach adapted in MIDS is to create a fingerprint of the image. The fingerprint is a 128-bit integer unique to that image created by an one-way hashing function. The calculation of fingerprint takes less than a fraction of seconds in our Unix workstations. The fingerprint can be either embedded in a selected attribute field of the image header or sent together as a separate data file. The receiving site can verify the authenticity of information received by calculating the fingerprint value of the received image and compare that with the received fingerprint. Any discrepancy between the two indicates the tampering of the image during the transmission. Note that the fingerprint method is a tradeoff between timing performance and data privacy. It detects unauthorized modification to the image data but does not prevent the image from unauthorized access.

5. CONCLUSIONS

Image database management with content-based retrieval will provide an environment for effective clinical decision making. It will open up many

opportunities in advancing medical services and reducing health care costs. There, however, lacks a clear definition of medical image database systems; let alone developing much needed tools that cannot be found in traditional databases.

The purpose of this paper is to present our design approach and implementation experience of image data warehouse at UCSF. We identified challenging issues and necessary tools for this new class of database systems. We provided a generic architecture of the vertically integrated medical image database system being developed at UCSF. We described the design methods and application-specific tools developed within this information framework and illustrated them with implemented examples.

Among the new database issues, the infrastructure support for medical data integration and content-based indexing of medical images, are the most immediate ones that must be first resolved. For the former issue, HI-PACS provides a genuine testbed for medical image database prototyping by removing the administrative and technological barriers of gathering information from diverse imaging and data sources. For the latter issue, the a priori approach is used to extract relevant features of PACS images and reports. These processed data then are translated and integrated into a common data model in the IDBS for enabling content-based information retrieval. The integration of content-based retrieval of relevant images and knowledge base techniques for decision making in the existing PACS installations will make the whole system much more useful in medical domain. Currently, tools for automatic mapping and correlating images of non-rigid body organs are still lacking.

In addition to current applications in epilepsy surgical planning, bone age assessment, and myelination disorders diagnosis, we will continue to identify and develop new MIDS applications driven by sound hypotheses. By gradually enlarging the scope of applications, we will learn more functionality for content-based query processing. The application-specific tools developed in these application can also be extracted and composed into a more generic package for medical image database development.

The research into medical image database systems has just begun. Many fundamental challenges in image information processing have yet to be resolved. Research must proceed in multidisciplinary teams that combine the expertise of database and imaging disciplines with the knowledge of the medical community. The successful implementation of medical image database systems in the health care

system will have a direct impact on the cost and quality of patient care, contributing to improved disease diagnosis, therapy treatment, biomedical knowledge, and access to health care.

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